

UMians EduGig: A Student Freelance Services Mobile Application



**In Partial Fulfillment of the Requirements
In CCE106/L – Applications Development and Emerging Technologies**

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Chapter 1

I. Introduction

Students at the University of Mindanao already hire each other for small paid jobs: a poster for an org event, a tutor before an exam, someone to proofread a thesis chapter. Today those deals happen in Facebook groups and group chats, where posts disappear within hours, nobody knows who is reliable, and nothing records what was agreed. UMians EduGig is a Flutter mobile application that gives UM students one place to offer and hire skill-based services. A student signs in with their UM Google account, publishes a service listing or browses the ones already there, places an order, chats with the other party, receives the work, and leaves a review. Every account can both buy and sell, so there is no separate freelancer registration. The app runs on Firebase (Authentication, Cloud Firestore, Cloud Storage, and Cloud Functions) and targets Android, with the same code running on the web and Windows for testing.

II. Purpose and Value Proposition

The purpose of UMians EduGig is to turn informal peer hiring into orders that both sides can trust without knowing each other first. The current process fails in four ways. Discovery is poor because posts scroll away and cannot be searched. Trust is weak because testimonials are screenshots, which are easy to fake. Structure is missing because the price, the deadline, and the number of revisions are never written down. Payment is awkward because either the buyer pays first and hopes, or the freelancer works first and hopes. Global platforms such as Upwork and Fiverr solve these for professionals, but their dollar pricing, worldwide competition, and payout requirements make no sense for a student charging 300 pesos for an hour of tutoring.

The app answers each of these gaps. Listings are searchable by category, keyword, and price. Reviews can only be written by the buyer of a completed order, one per order, so they cannot be faked or duplicated. Every order freezes its price, delivery days, revisions, and requirements at creation, and neither side can edit them afterwards. Payment goes through a Xendit checkout, and the platform holds the money until the client accepts the delivery, refunds it if the order is cancelled, or lets staff decide on a dispute. The platform keeps a 10 percent commission on completed orders and offers an optional Pro subscription for featured listings and a verified badge. What makes the app different is campus scale. The person you hire may sit beside you next semester, and the app adds a written record and a review trail on top of trust that already exists.

III. Target Audience

The users of UMians EduGig are students of the University of Mindanao, mostly college students aged 18 to 24. Only a verified umindanao.edu.ph Google account can sign in, so every user on the platform is a UM student, and the student number is read from the email address itself. Buying is open to everyone; selling and payouts require a birth date showing 18 or over. Each account can act in two roles. Buyers need short, deadline-bound tasks such as thesis proofreading, tutoring, posters for a student organization, video editing for a defense, or help with a programming assignment. They are price sensitive and already used to hiring peers informally, but they lack a quick way to find the right person and some assurance that the work will arrive. Sellers have a skill someone will pay for, such as design, programming, writing, video, photography, or music, and want income that fits between classes. They cannot commit to fixed hours and are not competitive on global platforms, but they are well matched to campus-scale work.

The app fits them by pricing in pesos, measuring delivery in days, and organizing listings into the eight categories students actually trade in: Tutoring, Graphic Design, Writing and Editing, Programming, Video Editing, Photography, Music and Audio, and Other. Profiles show each student's college and program from a closed list, so a buyer can see that a calculus tutor really is from the College of Engineering. A seller can pause a listing during exam week instead of deleting it, and released earnings are paid out to GCash, Maya, or a bank account.

IV. Core Functionalities

1. Marketplace with Search, Offers, and Ordering

The marketplace shows every published service as a card with its title, category, price, delivery time, seller, and rating. Buyers filter by category with a strip of ChoiceChip widgets, search with a TextField, and sort by newest or price. Results load through a ListView.builder on phones and a GridView on wider screens, paging with a Firestore cursor as the user scrolls. When a seller saves a listing, the app derives keyword tokens from the title and skills, so "poster" finds "Event poster for your student org". A listing is either fixed price, which can be ordered directly, or a starting price, where the buyer messages the seller and the seller sends an offer card in the chat with the agreed price, delivery days, and scope; accepting the card creates the order from those figures. This matters because a listing that cannot be found two weeks after posting means nothing else in the app gets used. Tapping a card pushes the service page through go_router, where the order or message button stays pinned at the bottom of the screen.

2. Order Lifecycle with Held Payment

An order records what was agreed and moves through pending, accepted, in progress, submitted, and completed, with a revision loop from submitted and with rejected, cancelled, and disputed as exits. On creation the price and terms are copied from the service inside a Firestore transaction, so a tampered request cannot change them. A transition table lists which state may move to which and which party may make each move: the seller accepts, starts, and submits; the buyer completes or asks for a revision. This is enforced three times. The screen only shows legal buttons, the repository re-checks inside a transaction, and Firestore security rules reject anything else. Once the seller accepts, the buyer pays through a Xendit checkout page and the platform holds the money until the order ends, then releases it to the seller's wallet minus commission, or refunds it. A client who ignores a delivery for three days does not keep the freelancer unpaid; a scheduled Cloud Function completes the order. The order screen listens to the document with a StreamBuilder, so when the other party acts, the status tracker and buttons update without a refresh. Cancelling or disputing opens an AlertDialog that asks for a reason first.

3. Chat, Reviews, and Notifications

A buyer and seller message each other from a service page or an order, with text, image, and file attachments stored in Cloud Storage. The conversation id is the two user ids sorted and joined, so a pair always lands in the same thread. A reversed ListView.builder keeps the newest message at the bottom, and the unread badge on the NavigationBar updates from the same stream. After completion the buyer leaves a star rating and comment through a dialog. The review's document id is the order id, so a second review on the same order is impossible, and the same atomic batch updates the service's rating counters. Every order and chat event writes an

inbox document for the recipient, which a Cloud Function fans out as a push notification through Firebase Cloud Messaging. Tapping a notification goes through one NotificationRouter that validates the id and opens the exact order or chat. These pieces matter because negotiation, delivery, and the evidence for a dispute all live in the chat, and reviews are the trust signal a screenshot cannot provide.

4. Admin Console with Role-Based Access

Staff run the platform from a console inside the same app. Access is the existence of an admins document for the signed-in student, written only with a service account key, and it carries a role and a list of permissions: manage students, moderate listings, manage categories, resolve disputes, settle payouts, handle refunds, decide identity verification, view reports, and edit platform settings. The screen builds one section per permission the staff member actually holds, as a NavigationRail on wide screens and a ChipStrip on phones, and each queue is a StreamBuilder over a ListView.builder with a Badge showing how many items are waiting. Disputes close as a full release or a full refund from an AlertDialog that records the reason, and orders of 2,000 pesos or more need one staff member to propose and a different one to confirm. Payouts are sent through Xendit or recorded by hand, stuck refunds are retried, chargebacks are passed on to the seller, and a student ID is approved only after the reviewer confirms it matches the declared name and birth date. Staff can also suspend an account, take down a listing, post an announcement, and pause new orders platform-wide. The same permission names are checked in the security rules and in the backend, the main admin grants staff but never themselves, and every staff action lands in an audit log that only the backend can write. This matters because money and identity decisions need a named person behind them and a record that cannot be edited afterwards.

V. Technical Feasibility & Flutter Elements

UMians EduGig is feasible with Flutter and Firebase because both are free to start, well documented, and already used in the course. The app is written in Dart 3 with Material 3 and uses `firebase_core`, `firebase_auth`, `cloud_firestore`, `firebase_storage`, and `firebase_messaging` for the backend, `go_router` for navigation, `provider` for state, `intl` for peso and date formatting, and `file_picker` and `url_launcher` for attachments and the checkout page. Code under `lib` is split by feature into `data`, `domain`, and `presentation` folders. Only repositories touch the Firebase SDKs; no widget imports Firestore. Repositories are created once at startup as plain Providers, and a single `AuthController` `ChangeNotifier` drives `go_router`'s redirect, so a signed-out or non-UM account cannot reach a protected route. A `ShellRoute` wraps the four tabs (Home, Orders, Messages, Me) in a `NavigationBar` on phones and a `NavigationRail` from 1000 pixels wide. Live data uses `StreamBuilder` with streams created once in `initState` and held, because a fresh stream on every rebuild tears down the Firestore listener. Forms use `Form` and `TextFormField` with validators kept in one file, and every Firebase error is mapped in the repository to a small failure class whose message is the only thing a screen shows.

Anything involving money or entitlements is decided on the server, never in the app, because an APK decompiles and a web build ships readable JavaScript. Cloud Functions in the `asia-southeast1` region hold the Xendit secret key, re-derive every amount from the order, and are the only writer of payment verification, wallet balances, payouts, and Pro subscriptions; Firestore rules make those fields unreachable from clients. A gateway callback is checked against a shared token in constant time and applied inside an idempotent transaction, and a scheduled job asks Xendit every thirty minutes about any checkout still pending, so a lost callback is a delay rather than a stuck order. A card chargeback after release is recorded by staff and taken from the seller's

balance, with any remainder repaid from their next releases. Disputes of 2,000 pesos or more need one staff member to propose the outcome and a different one to confirm it, and every staff action lands in an audit log only the backend can write.

Chapter 2

VI. Prototypes

Login and Account Selection

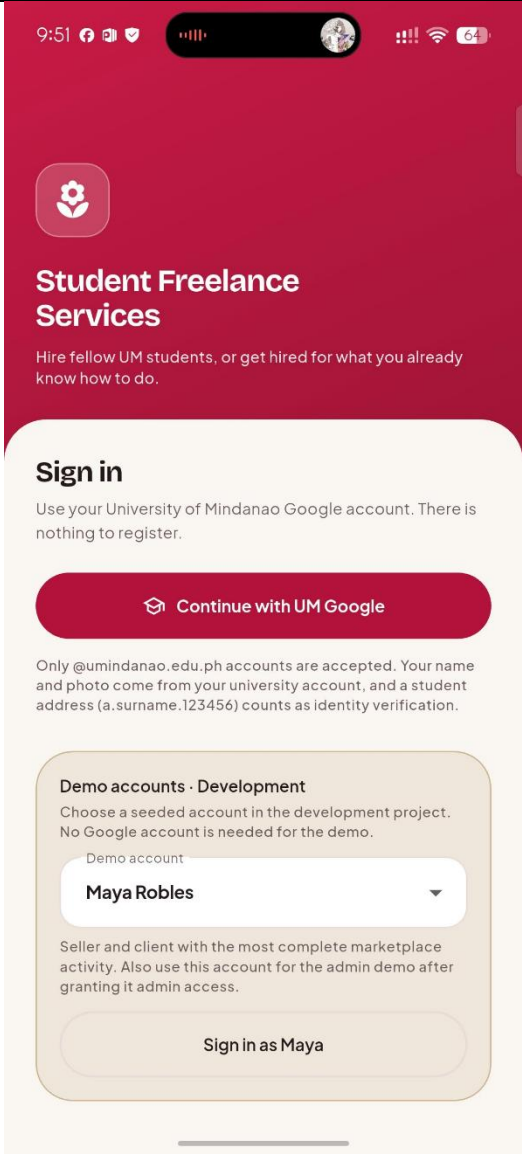
Login and Account Selection	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The entry screen provides UM Google sign-in and a development-only demo-account selector. Authentication prevents protected features from opening before an eligible profile is loaded. User interaction: The student chooses an account and signs in; the app loads the profile then routes the student to Home.	

Figure 1. Login and Account Selection

Marketplace Home and Service Cards

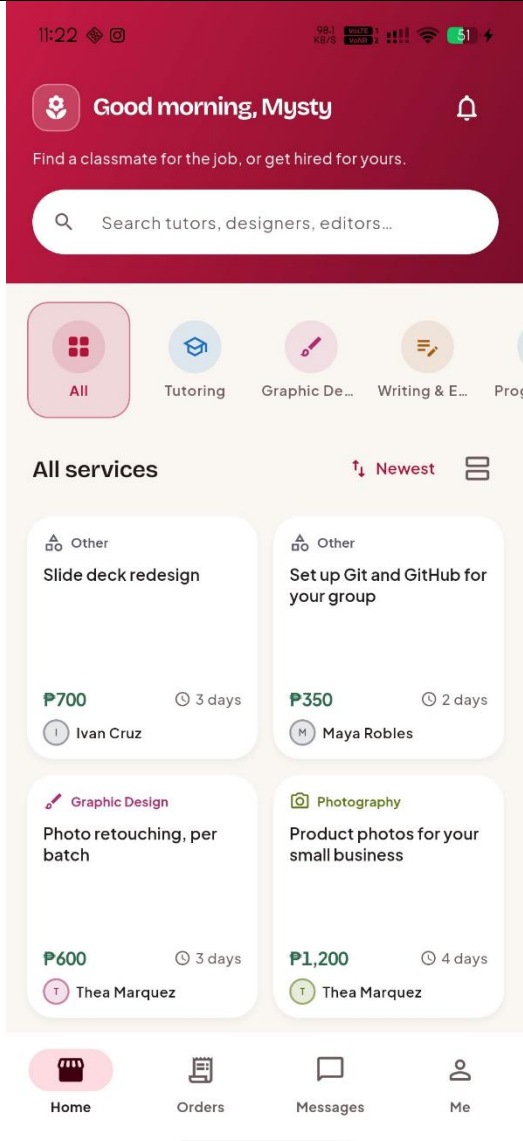
Marketplace Home and Service Cards																										
Explanation / Description	Mobile Perspective Screenshot																									
Purpose: Home combines search, categories, sorting, price, rating, seller identity, and delivery duration so clients can compare listings. At phone width, users can switch to a vertically scanned list. User interaction: The student searches or filters services, changes view mode if needed, and taps a listing to open it.	 The screenshot displays the mobile interface of the Marketplace Home. At the top, the status bar shows the time as 11:22. Below this, a header section features a greeting "Good morning, Mysty" and a search bar with the placeholder text "Search tutors, designers, editors...". A category bar follows, with "All" selected and other categories like "Tutoring", "Graphic De...", "Writing & E...", and "Pro" visible. The main content area is titled "All services" and includes a "Newest" sort option. It displays a grid of service cards. Each card contains a category icon, a title, a price, a delivery duration, and a seller name. <table><tr><th>Category</th><th>Service Title</th><th>Price</th><th>Duration</th><th>Seller</th></tr><tr><td>Other</td><td>Slide deck redesign</td><td>P700</td><td>3 days</td><td>Ivan Cruz</td></tr><tr><td>Other</td><td>Set up Git and GitHub for your group</td><td>P350</td><td>2 days</td><td>Maya Robles</td></tr><tr><td>Graphic Design</td><td>Photo retouching, per batch</td><td>P600</td><td>3 days</td><td>Thea Marquez</td></tr><tr><td>Photography</td><td>Product photos for your small business</td><td>P1,200</td><td>4 days</td><td>Thea Marquez</td></tr></table>	Category	Service Title	Price	Duration	Seller	Other	Slide deck redesign	P700	3 days	Ivan Cruz	Other	Set up Git and GitHub for your group	P350	2 days	Maya Robles	Graphic Design	Photo retouching, per batch	P600	3 days	Thea Marquez	Photography	Product photos for your small business	P1,200	4 days	Thea Marquez
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Figure 2. Marketplace Home and Service Cards

Service Detail

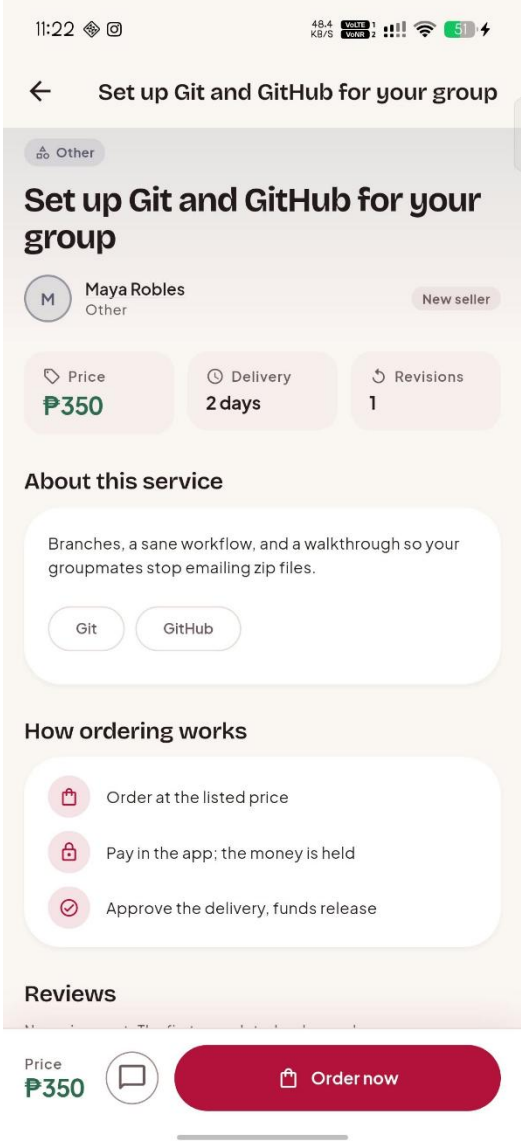
Service Detail	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The listing detail explains seller, price, delivery time, revisions, description, and reviews before the client commits. User interaction: A client orders fixed-price work or messages the seller to negotiate.	

Figure 3. Service Detail

Create Order

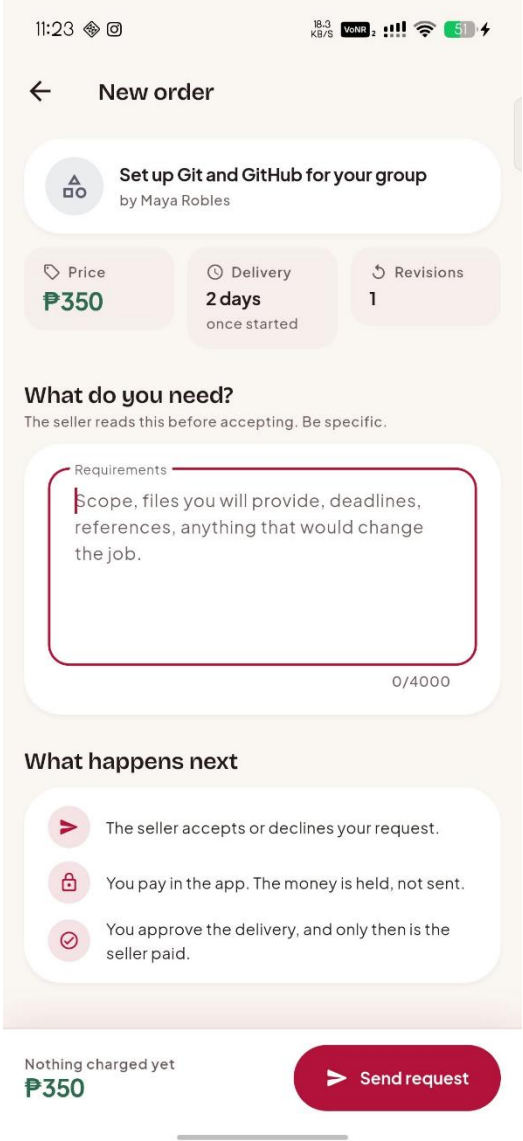
Create Order	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The order form captures requirements while displaying terms that will be copied and frozen on the order record. User interaction: The client enters a brief, passes validation, confirms the order, and begins payment.	

Figure 4. Create Order

Chat and Offer Negotiation

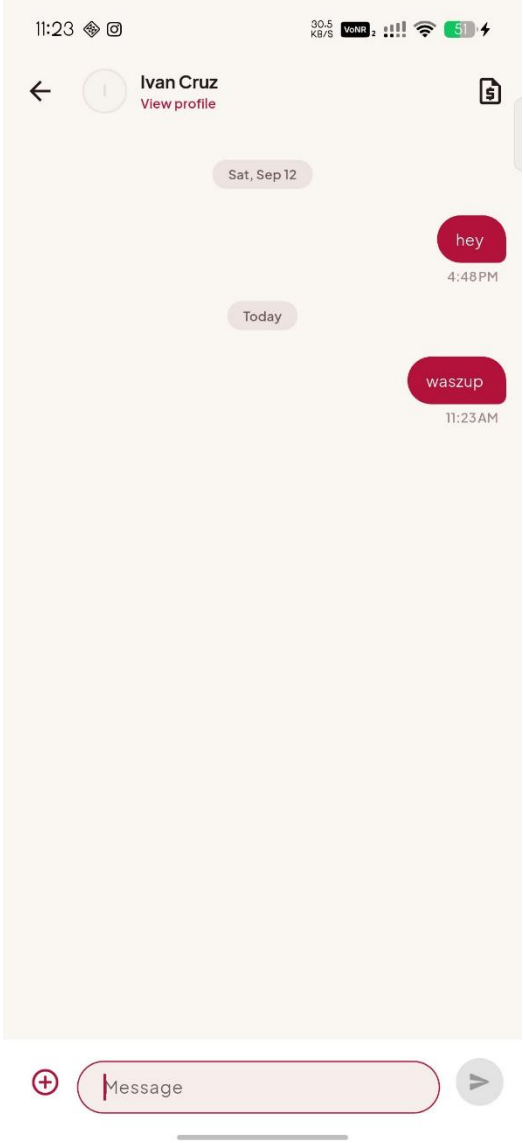
Chat and Offer Negotiation	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The conversation keeps negotiation, messages, files, and offer cards together as written evidence of an agreement. User interaction: The freelancer sends an offer; the client reviews, accepts, or declines it.	 The screenshot shows a mobile chat interface. At the top, the status bar displays the time 11:23, signal strength, VoWiFi, and battery level. The chat header shows a back arrow, a profile icon, the name 'Ivan Cruz', and a 'View profile' link. A date separator indicates 'Sat, Sep 12'. The chat history includes a red bubble with the text 'hey' at 4:48 PM and another red bubble with 'waszup' at 11:23 AM. A 'Today' separator is also visible. At the bottom, there is a text input field with a plus icon on the left, a 'Message' placeholder, and a send button on the right.

Figure 5. Chat and Offer Negotiation

Orders List

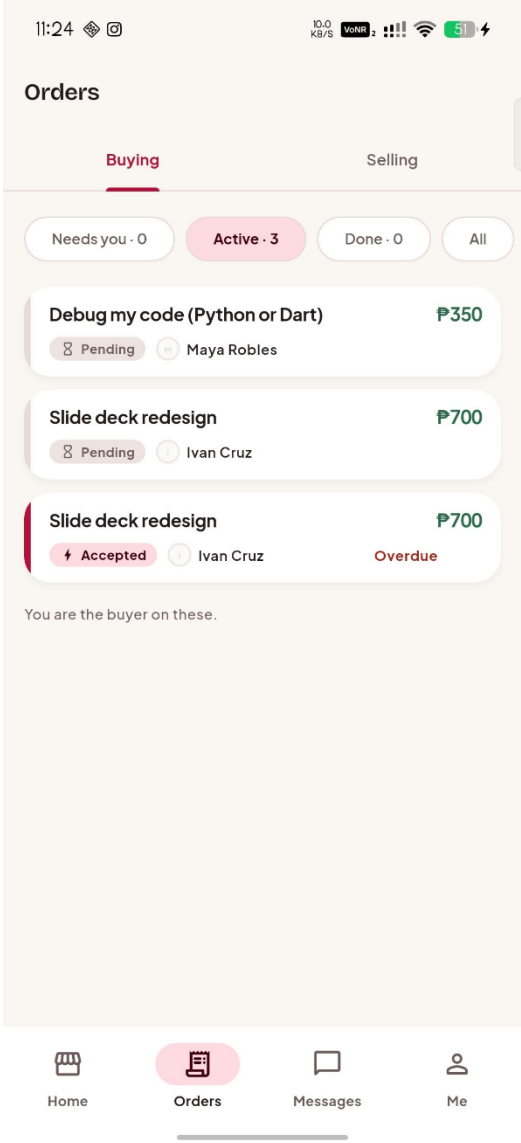
Orders List	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The Orders tab presents live jobs with their current state and relevant due information for the signed-in participant. User interaction: The client or freelancer opens an order to see terms, payment state, and legal next actions.	

Figure 6. Orders List

Order Detail and Lifecycle

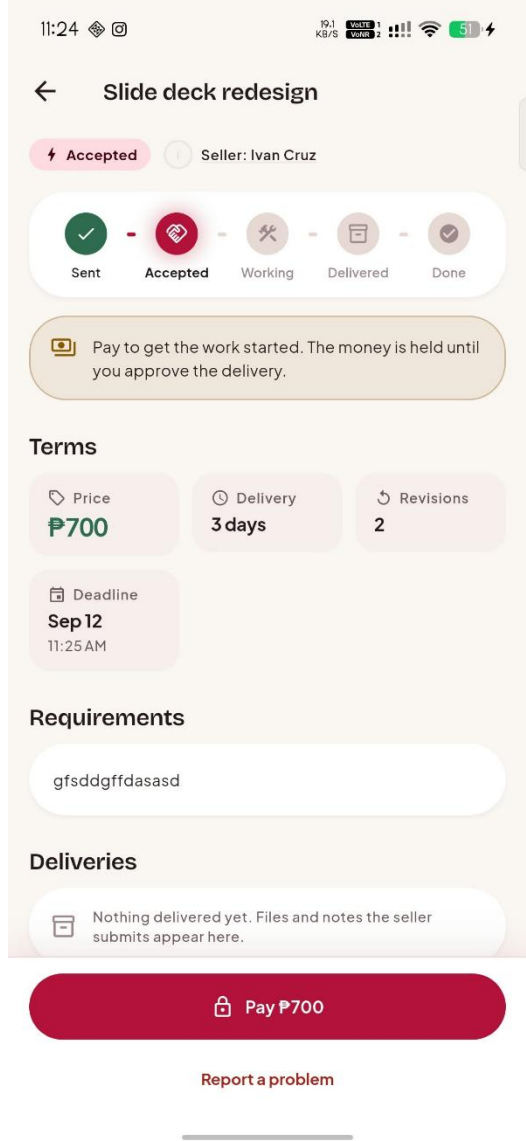
Order Detail and Lifecycle	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The detail page combines the status tracker, agreed terms, delivery evidence, and a persistent action area that changes by role and state. User interaction: The freelancer accepts, starts, or submits work; the client pays, revises, accepts, cancels, or disputes.	 The screenshot shows a mobile app interface for an order titled "Slide deck redesign". At the top, the status is "Accepted" with a lightning bolt icon, and the seller is "Ivan Cruz". Below this is a progress bar with five stages: "Sent" (green checkmark), "Accepted" (red circle with white checkmark, currently active), "Working" (grey circle with white X), "Delivered" (grey circle with white document icon), and "Done" (grey circle with white checkmark). A yellow box with a wallet icon states: "Pay to get the work started. The money is held until you approve the delivery." Under the "Terms" section, there are three cards: "Price" showing "P700", "Delivery" showing "3 days", and "Revisions" showing "2". A "Deadline" card shows "Sep 12" at "11:25 AM". The "Requirements" section contains a text box with the placeholder "gfsddgffdasasd". The "Deliveries" section shows a message: "Nothing delivered yet. Files and notes the seller submits appear here." At the bottom, there is a large red button labeled "Pay P700" and a smaller link "Report a problem".

Figure 7. Order Detail and Lifecycle

Payment Confirmation

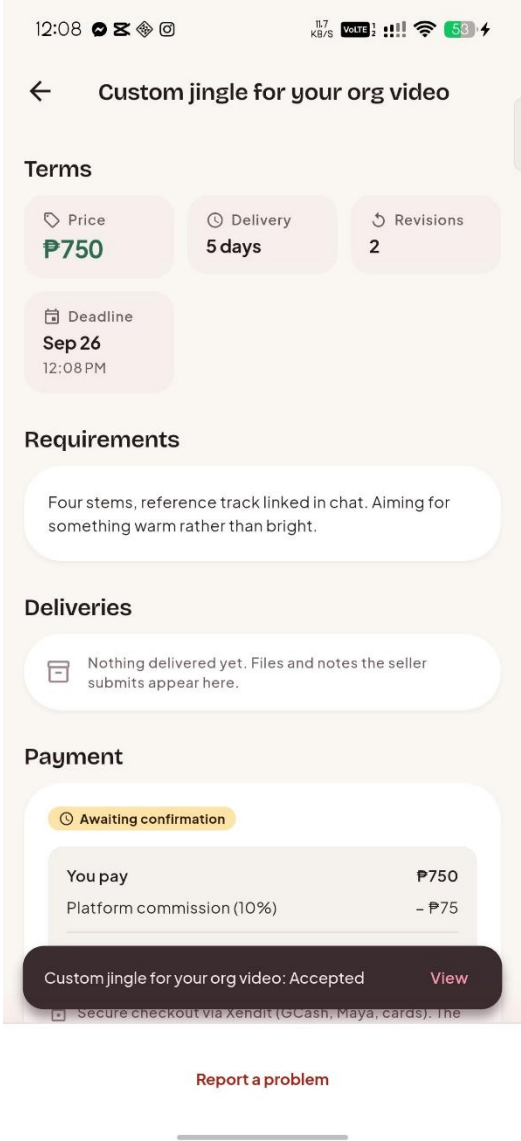
Payment Confirmation	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The payment interface explains that funds are held until completion or an authorized refund/dispute outcome. User interaction: The client chooses a method or submits a manual reference; verification updates the order.	

Figure 8. Payment Confirmation

Student Profile Hub

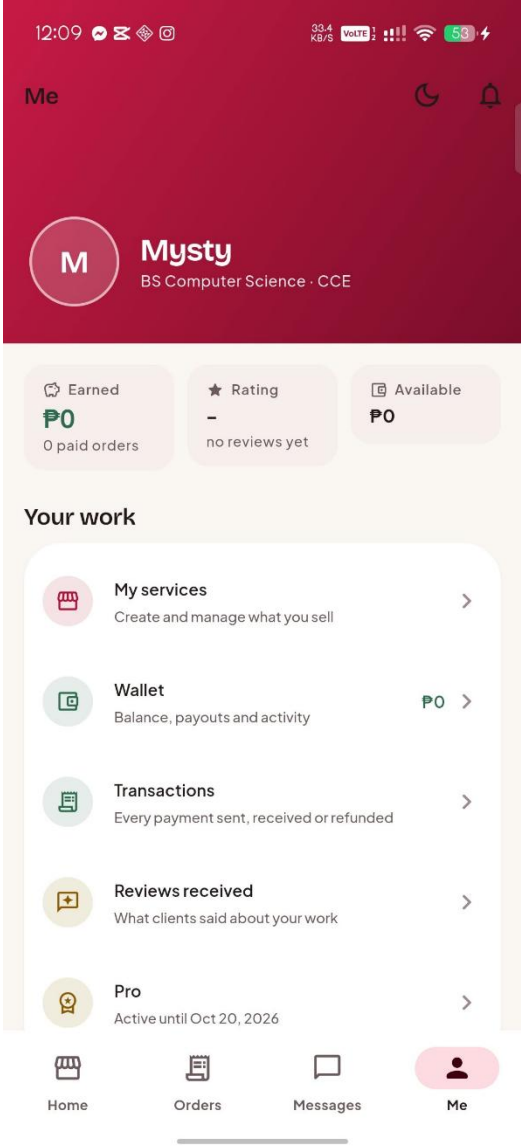
Student Profile Hub	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The profile hub links public information, services, wallet, Pro option, verification, and notifications. User interaction: The student selects a profile-management destination and changes information they own.	 The screenshot shows a mobile app interface for a user named 'Mysty' (BS Computer Science - CCE). The top status bar shows the time 12:09, signal strength, and battery level at 53%. The app header is red with the user's name 'Me' and a profile icon 'M'. Below the header, there are three status boxes: 'Earned PO' (0 paid orders), 'Rating' (no reviews yet), and 'Available PO'. The main section is titled 'Your work' and contains a list of items: 'My services' (Create and manage what you sell), 'Wallet' (Balance, payouts and activity) with a 'PO' badge, 'Transactions' (Every payment sent, received or refunded), 'Reviews received' (What clients said about your work), and 'Pro' (Active until Oct 20, 2026). The bottom navigation bar has four icons: Home, Orders, Messages, and Me (which is highlighted).

Figure 9. Student Profile Hub

Public Student Profile

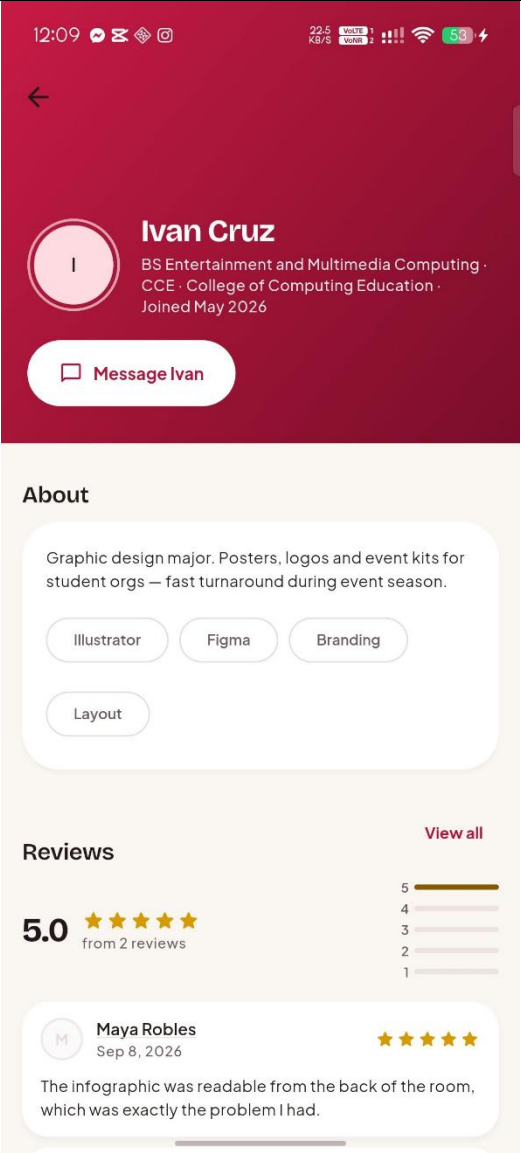
Public Student Profile	
Explanation / Description	Mobile Perspective Screenshot
Purpose: A public profile provides college, programme, listings, and completed-order reviews to support informed hiring. User interaction: A client views listings or reviews and starts a contextual conversation with the freelancer.	 The screenshot shows a mobile app interface for a student profile. At the top, the status bar shows the time 12:09, signal strength, and battery level at 53%. The profile header has a dark red background with a back arrow, a circular profile picture placeholder with the letter 'I', and the name 'Ivan Cruz'. Below the name, it says 'BS Entertainment and Multimedia Computing · CCE · College of Computing Education · Joined May 2026'. A 'Message Ivan' button is present. The 'About' section describes Ivan as a 'Graphic design major' who creates 'Posters, logos and event kits for student orgs'. It lists skills: 'Illustrator', 'Figma', 'Branding', and 'Layout'. The 'Reviews' section shows a 5.0 rating from 2 reviews, with a 'View all' link. A review from 'Maya Robles' dated 'Sep 8, 2026' is shown, stating 'The infographic was readable from the back of the room, which was exactly the problem I had.' and is accompanied by five stars.

Figure 10. Public Student Profile

My Services

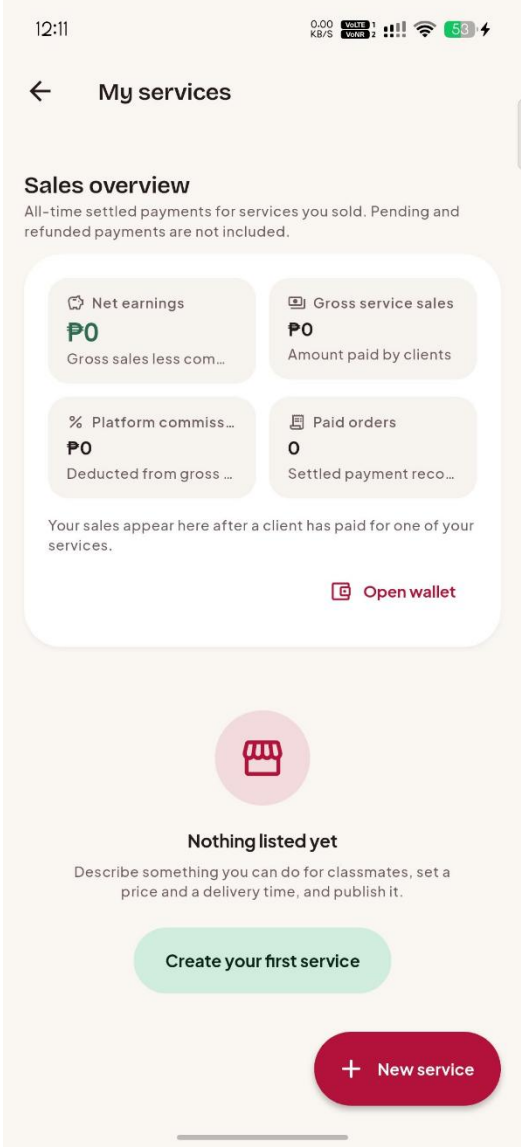
My Services	
Explanation / Description	Mobile Perspective Screenshot
Purpose: Freelancers see draft, published, paused, and archived listings without exposing unpublished work to clients. User interaction: The owner creates, edits, publishes, pauses, or archives their services.	

Figure 11. My Services

Create Service Listing

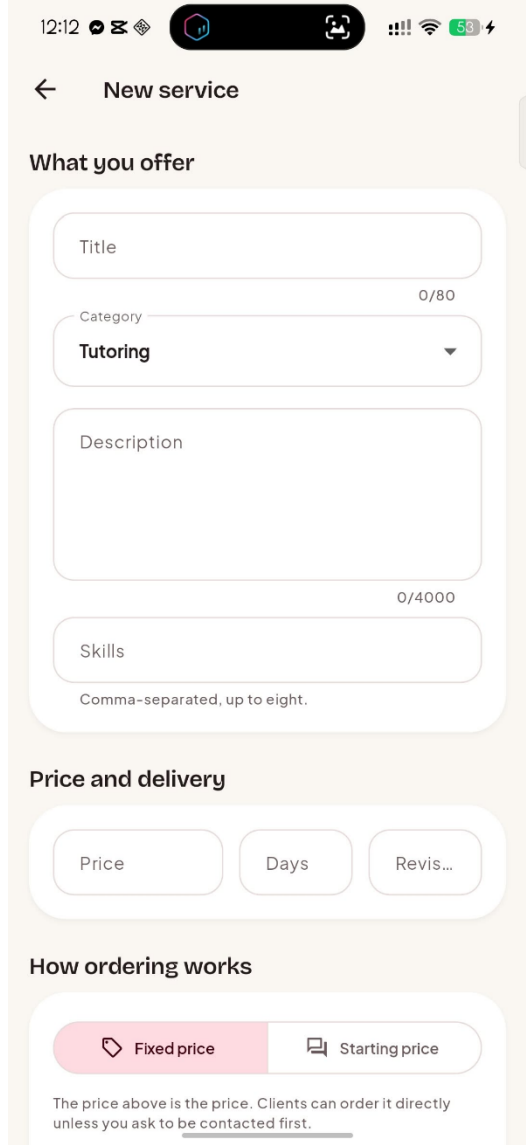
Create Service Listing	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The form validates title, category, skills, price, duration, revisions, and order method before publishing. User interaction: The freelancer enters terms, corrects messages, saves a draft, or publishes.	 The screenshot shows a mobile app interface for creating a new service. At the top, there's a status bar with the time 12:12 and various icons. Below it, a back arrow and the title 'New service' are visible. The form is divided into three main sections: 'What you offer', 'Price and delivery', and 'How ordering works'. The 'What you offer' section contains a 'Title' field (0/80), a 'Category' dropdown menu (currently set to 'Tutoring'), a 'Description' field (0/4000), and a 'Skills' field (comma-separated, up to eight). The 'Price and delivery' section has three input fields: 'Price', 'Days', and 'Revis...'. The 'How ordering works' section features two radio buttons: 'Fixed price' (selected) and 'Starting price'. Below these, a note states: 'The price above is the price. Clients can order it directly unless you ask to be contacted first.'

Figure 12. Create Service Listing

Wallet

Wallet	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The wallet separates held, clearing, available, and payout-requested funds so sellers understand balance movement. User interaction: A qualified freelancer opens payout setup or requests a payout from available funds.	

Figure 13. Wallet

Payout Account Setup

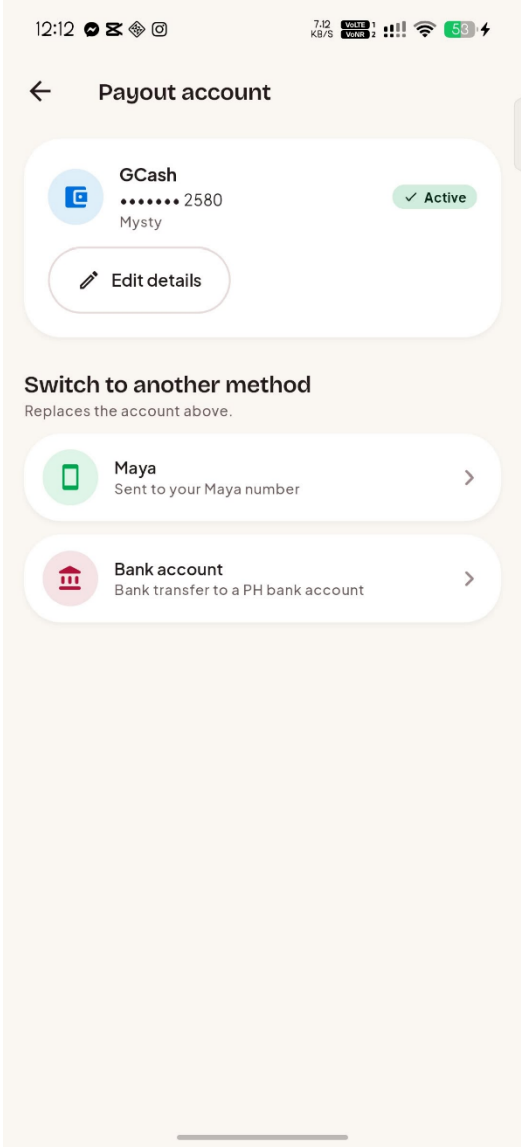
Payout Account Setup	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The payout form validates supported GCash, Maya, or bank details before an account becomes a payment destination. User interaction: The seller chooses a provider, enters details, and saves it for future requests.	 The screenshot shows a mobile application interface for setting up a payout account. At the top, the status bar displays the time 12:12, various notification icons, and network information (7.12 KB/s, 5G). The app header is titled 'Payout account' with a back arrow. The main content area features a 'GCash' entry with a blue icon, a masked card number ending in 2580, the name 'Mysty', and a green 'Active' status indicator. Below this is an 'Edit details' button. A section titled 'Switch to another method' with the subtitle 'Replaces the account above.' offers two alternative options: 'Maya' (with a green icon and the instruction 'Sent to your Maya number') and 'Bank account' (with a red icon and the instruction 'Bank transfer to a PH bank account'). Both options have right-pointing chevron arrows.

Figure 14. Payout Account Setup

Wallet Transactions

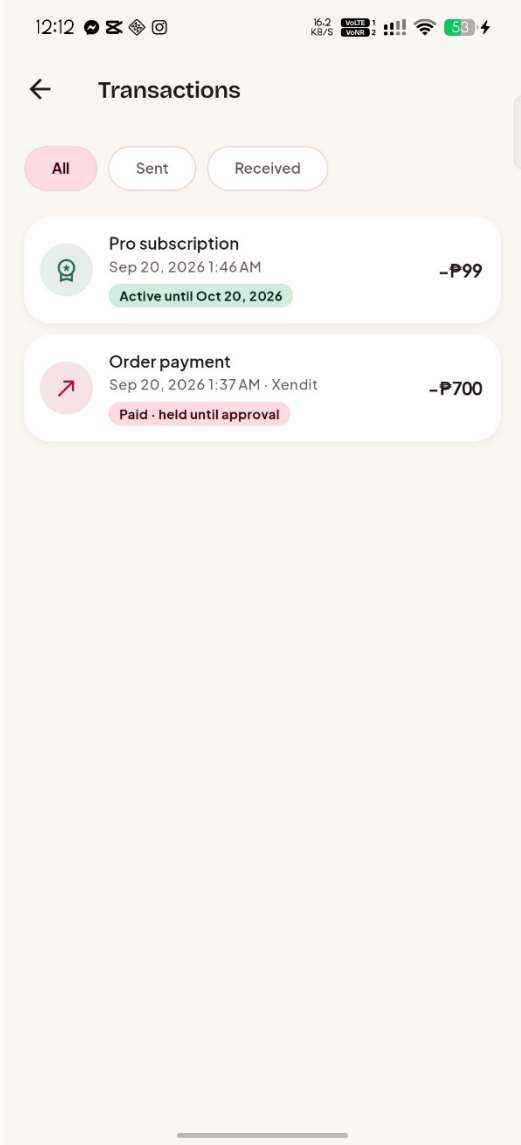
Wallet Transactions	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The ledger records releases, clearance, payout requests, refunds, and chargeback adjustments. User interaction: The seller reviews entries to reconcile the available wallet balance.	

Figure 15. Wallet Transactions

Pro Subscription and Verification

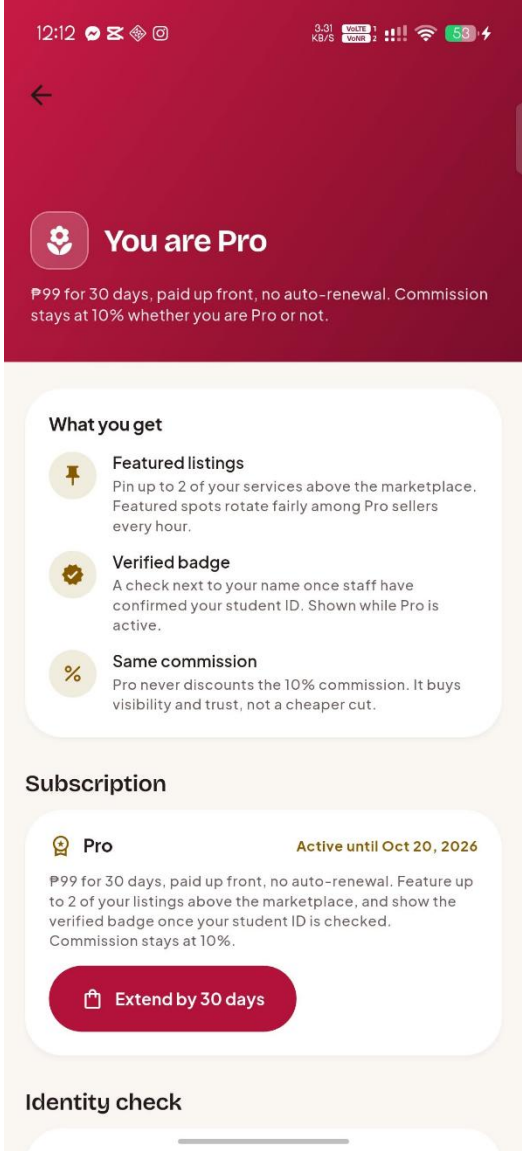
Pro Subscription and Verification	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The Pro screen explains the optional subscription, featured-listing allowance, and verified-badge requirement. User interaction: The student begins a subscription payment or submits identity evidence for staff review.	

Figure 16. Pro Subscription and Verification

Notifications

Notifications	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The inbox gathers order, chat, review, payout, and platform events in one place. User interaction: Tapping a notice validates its destination and opens the relevant task.	

Figure 17. Notifications

Conversations List

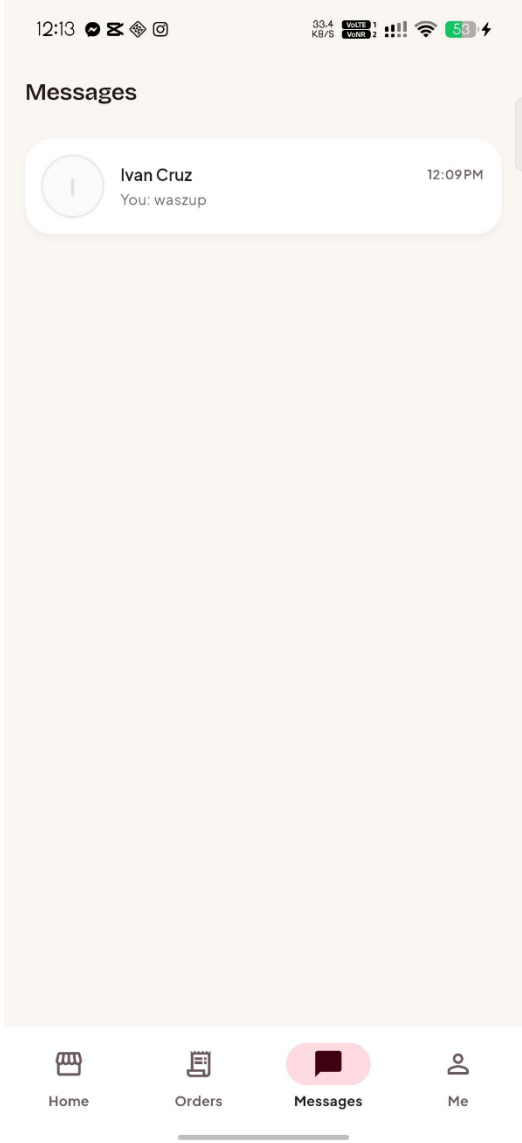
Conversations List	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The Messages tab shows conversations, recent activity, and unread state so students return to active work quickly. User interaction: The student opens a conversation to continue negotiation, send evidence, or answer a client.	 A mobile app screenshot showing the 'Messages' tab. The status bar at the top displays the time 12:13, various notification icons, and network status (0.3.4 KB/S, 5G). The main content area is titled 'Messages' and shows a conversation with 'Ivan Cruz' where the user sent 'waszup' at 12:09 PM. The bottom navigation bar has four icons: Home, Orders, Messages (which is highlighted with a red circle), and Me.

Figure 18. Conversations List

Reviews

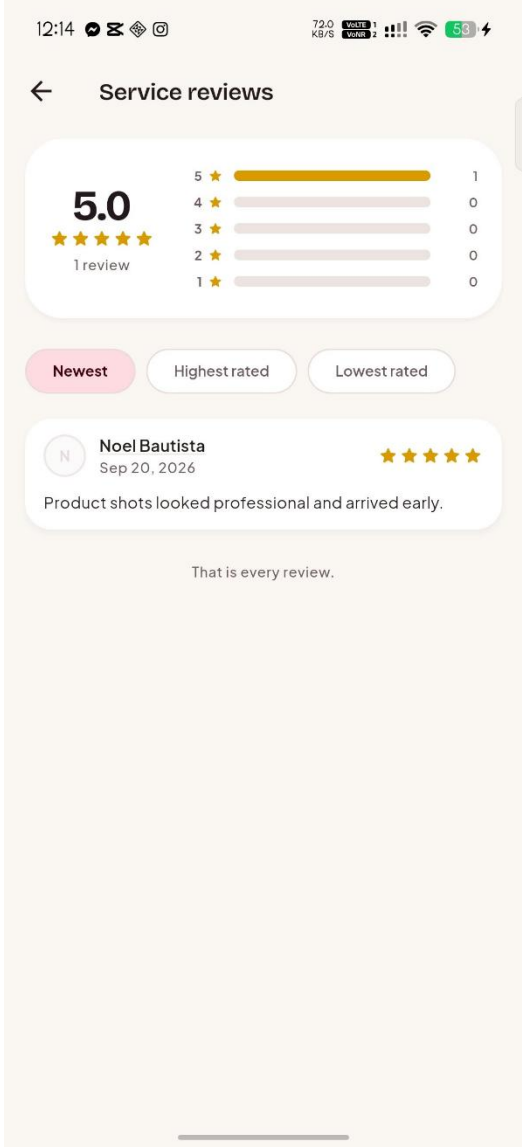
Reviews	
Explanation / Description	Mobile Perspective Screenshot
Purpose: Review data is tied to completed orders, preventing arbitrary or duplicate testimonials. User interaction: The client submits one rating after completion; users read feedback while evaluating a service.	 The screenshot displays a mobile application interface for viewing service reviews. At the top, the status bar shows the time as 12:14 and various connectivity icons. The page title is 'Service reviews' with a back arrow. The main content area features a large '5.0' rating with five stars and '1 review' text. To the right, a vertical bar shows the distribution of ratings: 5 stars (1), 4 stars (0), 3 stars (0), 2 stars (0), and 1 star (0). Below this, there are three sorting buttons: 'Newest' (selected), 'Highest rated', and 'Lowest rated'. A single review is shown by 'Noel Bautista' on 'Sep 20, 2026', with a five-star rating and the text 'Product shots looked professional and arrived early.' At the bottom, a placeholder text reads 'That is every review.'

Figure 19. Reviews

Role-Based Admin Console

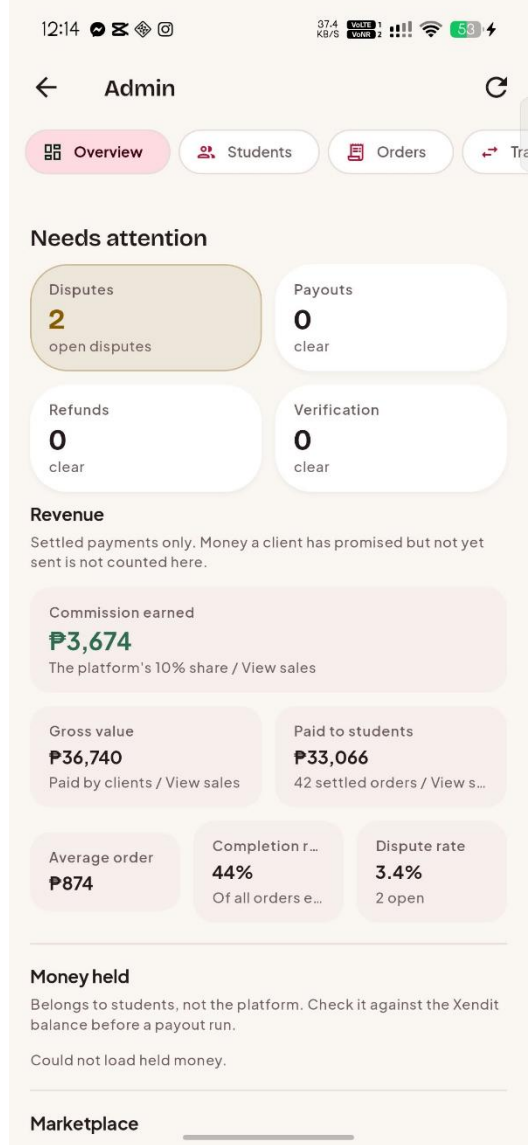
Role-Based Admin Console	
Explanation / Description	Mobile Perspective Screenshot
Purpose: The console centralizes queues and reports while limiting staff to permissions they were granted. User interaction: Authorized staff open profiles, moderation queues, orders, disputes, payouts, refunds, verification, sales, and audit details.	 The screenshot shows a mobile app interface for an 'Admin' console. At the top, there's a navigation bar with a back arrow, the title 'Admin', and a refresh icon. Below this is a horizontal menu with tabs: 'Overview' (selected), 'Students', 'Orders', and 'Transactions'. The main content area is divided into several sections: Needs attention: A section with four cards showing counts and 'clear' buttons: - Disputes: 2 open disputes - Payouts: 0 clear - Refunds: 0 clear - Verification: 0 clear Revenue: A section with a title 'Commission earned' showing a value of ₱3,674 and a note 'The platform's 10% share / View sales'. Gross value: A card showing ₱36,740 with the note 'Paid by clients / View sales'. Paid to students: A card showing ₱33,066 with the note '42 settled orders / View s...'. Average order: A card showing ₱874. Completion r...: A card showing 44% with the note 'Of all orders e...'. Dispute rate: A card showing 3.4% with the note '2 open'. Money held: A section with a title 'Money held' and a note 'Belongs to students, not the platform. Check it against the Xendit balance before a payout run.' Below this is a message 'Could not load held money.' Marketplace: A section at the bottom with a title 'Marketplace' and a horizontal bar.

Figure 20. Role-Based Admin Console

VII. Task Analysis/User Interface Manual

The manual explains the main transactions in the order real users perform them. Each visible action is reinforced by repository validation, security rules, and trusted backend checks where money or staff authority is involved.

A. Client: Find and Order a Fixed-Price Service

Step 1. Open Home. The marketplace loads published listings with category chips, search, sorting, and the list/card-view control.

Step 2. Enter a term or choose a category. The query narrows cards to listings matching the entered keyword and category.

Step 3. Open the selected listing. The detail page presents seller, price, duration, revisions, reviews, and the available ordering method.

Step 4. Select Order, write requirements, and submit the validated form. The system copies listing terms to a pending order so later listing changes cannot alter the agreement.

Step 5. Choose payment and complete checkout. The backend verifies payment information before fulfillment becomes eligible.

Step 6. Open Orders to monitor the lifecycle, accept delivery, request revision, cancel where legal, or open a dispute.

B. Client and Freelancer: Negotiate an Offer

Step 1. The client opens a negotiated listing and selects Message to get an offer, opening the conversation shared by the two students.

Step 2. Both parties discuss scope, cost, revisions, and timing in chat. Messages and attached files remain available evidence for a later dispute.

Step 3. The freelancer selects Send offer and enters the final scope, price, delivery days, and revisions. The chat receives an expiring offer card.

Step 4. The client reviews and accepts the offer. The application creates exactly one order using the offer values instead of the original starting price.

Step 5. The client pays and both participants continue through the normal fulfillment lifecycle.

C. Freelancer: Publish, Deliver, and Request Payout

Step 1. Open Me and select My Services to view personal draft, published, paused, and archived listings.

Step 2. Select Add service or edit an existing service. The form requires valid marketplace terms before publication.

Step 3. After a client pays, open Orders and accept the job. Start work, use chat for clarification, and submit delivery evidence when finished.

Step 4. When the client accepts, the backend applies commission and moves the net amount to the wallet.

Step 5. Open Wallet, configure a valid payout destination if needed, and request payout from available funds. The request enters the guarded settlement flow and ledger.

D. Authorized Staff: Moderate and Resolve

Step 1. Open Admin. Navigation is built from the staff member's permission list, so unauthorized queues are not usable work surfaces.

Step 2. Choose an authorized queue such as Students, Orders, Disputes, Payouts, Refunds, Verification, Sales, or Audit Log and inspect the relevant evidence.

Step 3. For student moderation, open the student profile and take down a listing directly. For financial and identity queues, select the permitted decision and supply required notes.

Step 4. Confirm the action. Protected money and permission operations go to the backend rather than direct client writes.

Step 5. Review the changed queue state and audit record, whose references resolve into readable student, service, order, offer, payment, or payout names.

VIII. Use Case

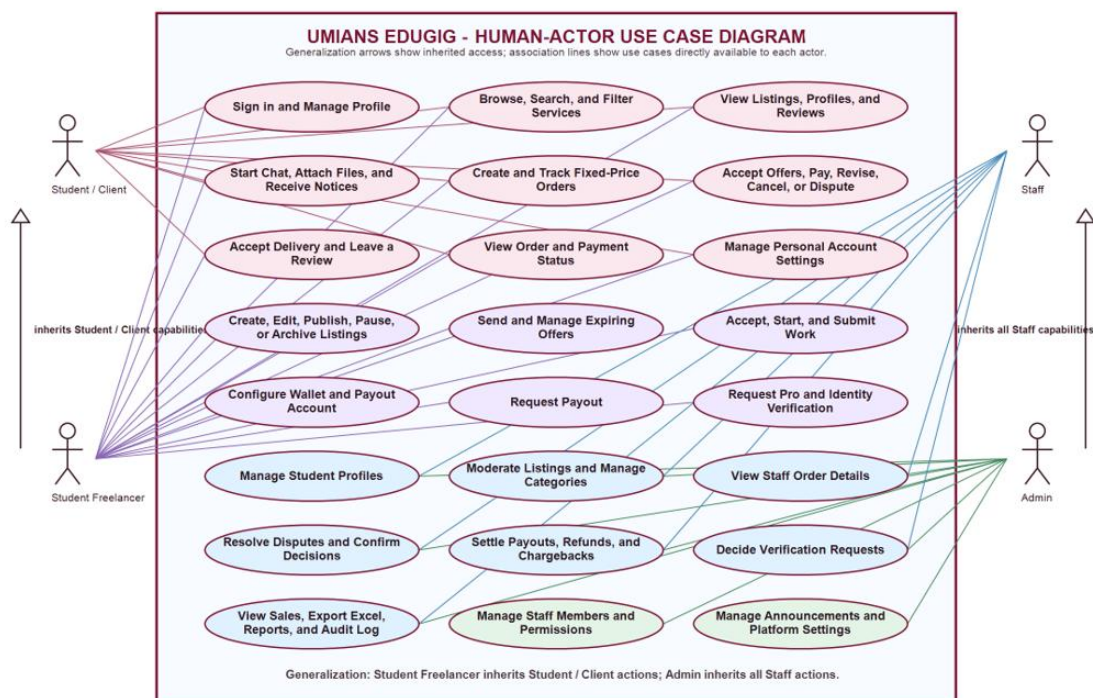


Figure 21. Complete human-actor use-case diagram for UMians EduGig

Student/Client covers account access, marketplace discovery, profiles, chat, fixed and negotiated orders, payment completion, tracking, revisions, cancellation, disputes, delivery acceptance, and reviews. The Student Freelancer has all of those actions and adds listings, offers,

work submission, wallet, payout, Pro, and verification. Staff actions remain permission-scoped. Admin inherits the entire Staff set such as student moderation, queues, financial decisions, verification, sales, Excel export, reports, and audit access, then adds staff-permission, announcement, and platform-setting management.

IX. Conceptual Framework

Figure 22 presents the Input-Process-Output framework. Student and staff/admin inputs pass through role-specific processes to produce marketplace, transaction, and management outputs.

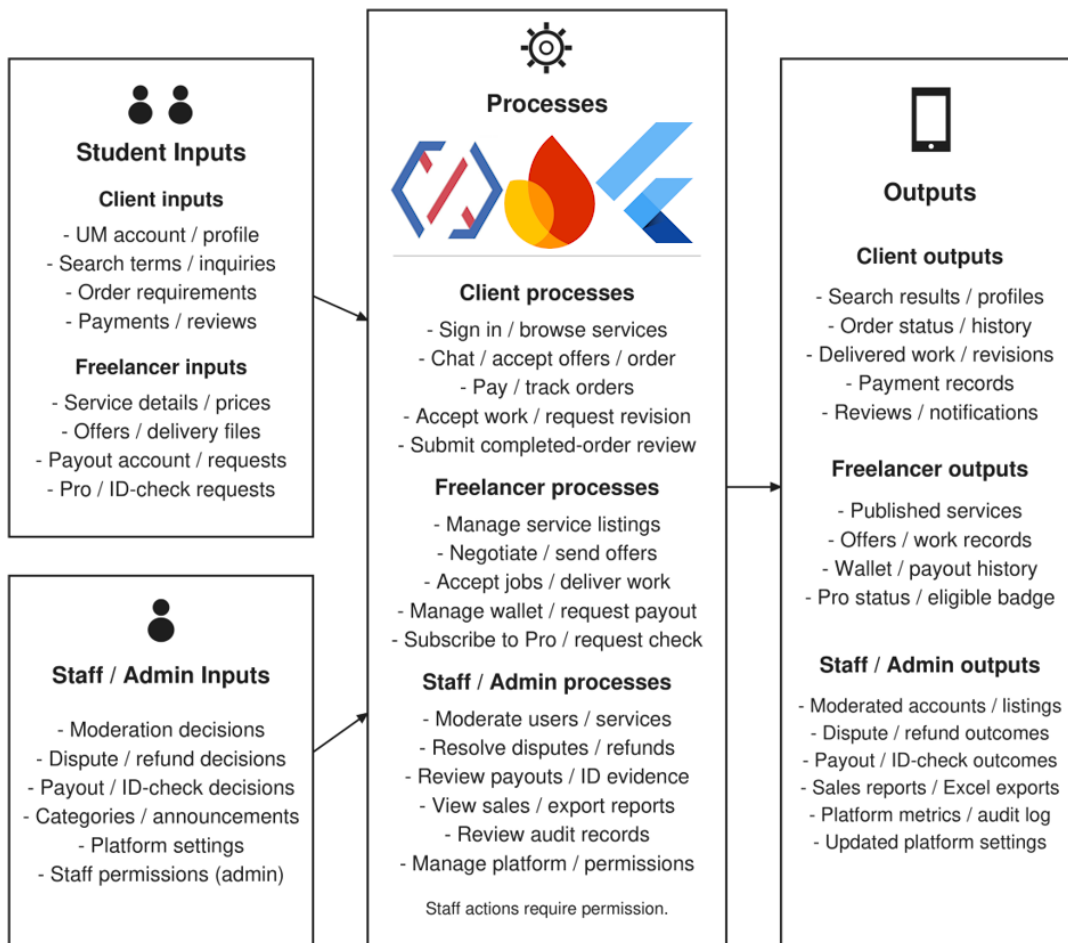


Figure 22. Input-Process-Output conceptual framework and tools used

Flutter and Dart provide the interface; Firebase supports authentication, data, files, notifications, and backend processing; Xendit handles gateway payments. Reviews and audit records inform later decisions.

X. Gantt Chart

Activity	Description	Responsibility	Dependency	W 4	W 5	W 6	W 7	W 8
Requirements refinement and UI prototype	Define system requirements and design screen layouts for the student marketplace.	April Bords Nerosa - Systems Analyst / UI Designer	—					
Authentication and profile foundation	Implement UM account sign-in, profile setup, and access to protected screens.	April Bords Nerosa - App Developer	Requirements					
Marketplace and service management	Build service search, filtering, listing creation, editing, and publishing.	April Bords Nerosa - App Developer	Authentication					
Chat, offers, and order lifecycle	Implement messaging, negotiated offers, order tracking, deliveries, and revisions.	April Bords Nerosa - App Developer	Marketplace					
Payments, wallet, and notifications	Integrate payment processing, wallet records, payout requests, and notifications.	April Bords Nerosa - Backend Developer	Orders					
Admin console, reports, and audit	Develop permission-based administration, moderation queues, sales reports, and audit records.	April Bords Nerosa - Full-stack Developer	Authentication; payments					
Integration, regression, and mobile UI testing	Test connected workflows, access controls, regressions, and mobile screen behavior.	April Bords Nerosa - Software Tester	All modules					
Chapter documentation and final presentation	Prepare chapter documentation, compile test evidence, and present the completed system.	April Bords Nerosa - Technical Writer / Presenter	Test evidence					

XI. Test Case

A. White-Box Test Cases — Client

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
WB-C-01	Search keyword derivation	Save a published listing, then search by title and skill token.	Only matching published listings are returned.	Search returned only published listings matching the title or skill token.	Passed
WB-C-02	Order-state guard	Attempt legal and illegal state transitions.	Only the authorized actor and legal next state succeed.	Authorized legal transitions succeeded; unauthorized and illegal transitions were rejected.	Passed
WB-C-03	Review uniqueness	Submit then duplicate a completed-order review.	Deterministic review id prevents a duplicate.	The first review was saved; the duplicate review was rejected.	Passed
WB-C-04	Marketplace pagination	Load a first query page, retain its cursor, and request the next page.	The next page adds only later matching listings with no duplicate record.	The next page returned later matching listings without duplicate records.	Passed
WB-C-05	Profile access guard	Attempt to open a protected client route while signed out and while signed in.	Signed-out navigation redirects to login; authenticated navigation opens the route.	Signed-out access redirected to login; signed-in access opened the protected route.	Passed
WB-C-06	Category filter combination	Combine a category selection, text query, and sort order.	The query returns only records satisfying every selected constraint.	Results matched the selected category and search term in the requested sort order.	Passed
WB-C-07	Notification target validation	Open notifications with valid, missing, and malformed target identifiers.	Only valid targets navigate; invalid targets show a safe failure message.	Valid targets opened; missing or malformed targets displayed a safe failure message.	Passed

WB-C-08	Suspended-user guard	Attempt protected client actions after a profile is suspended.	The account cannot continue protected marketplace activity.	The suspended account was prevented from continuing protected marketplace actions.	Passed
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B. White-Box Test Cases — Freelancer

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
WB-F-01	Listing ownership	Edit a listing as owner then as another user.	Only owner can change listing fields.	The owner could edit the listing; another user could not change its fields.	Passed
WB-F-02	Payout validation	Submit valid and invalid provider account formats.	Only valid payout details are accepted.	Valid payout details were accepted; invalid account formats were rejected.	Passed
WB-F-03	Delivery transition	Submit delivery before and after in-progress state.	Delivery is accepted only in permitted state.	Delivery was rejected before the permitted work state and accepted after work started.	Passed
WB-F-04	Listing field validation	Save listings with delivery, price, revision, and title values at valid and invalid boundaries.	Only values inside the defined validation range persist.	Valid listing values were saved; values outside the allowed limits were rejected.	Passed
WB-F-05	Offer expiry guard	Attempt to accept an offer before and after its expiry timestamp.	Only a still-valid pending offer can create an order.	A valid pending offer created an order; an expired offer was rejected.	Passed
WB-F-06	Draft visibility	Save a listing as draft and query marketplace results.	Draft data remains visible only to its owner.	The owner could view the draft; it did not appear in public marketplace results.	Passed
WB-F-07	Minimum payout guard	Request a payout below and at the configured minimum balance.	Below-minimum request is refused;	The below-minimum payout request was	Passed

			qualifying request is accepted.	rejected; the qualifying request was accepted.	
WB-F-08	Conversation participant guard	Attempt message or offer creation as a conversation participant and outsider.	Only participants can write to the conversation.	Conversation participants could send messages and offers; outsiders were denied.	Passed

C. White-Box Test Cases — Authorized Staff

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
WB-S-01	Permission boundary	Use an account lacking a queue permission.	Rules and backend reject access.	The account without the required permission was denied queue access.	Passed
WB-S-02	Moderation audit	Take down a listing as authorized staff.	Listing update and readable audit entry are recorded.	The listing was taken down and a readable audit entry recorded the action.	Passed
WB-S-03	Dispute confirmation	Confirm a qualifying decision as same then separate staff.	Same-person confirmation fails; independent confirmation succeeds.	Confirmation by the same staff member was rejected; independent confirmation succeeded.	Passed
WB-S-04	Sales-report scope	Query payment reports using all-time and date-range filters.	Only settled payments inside the selected range are returned.	Reports returned only settled payments within the selected date range.	Passed
WB-S-05	Audit reference resolution	Open entries targeting a user, service, order, payment, offer, and payout.	The display resolves readable subjects without exposing raw target ids.	Audit references displayed readable subjects for each tested record type.	Passed
WB-S-06	Verification permission	Attempt verification decisions with and without verification permission.	Only the authorized staff member can decide the request.	Authorized staff could decide verification requests; staff without permission were denied.	Passed
WB-S-	Refund permission	Attempt refund retry and manual	Only staff with refund	Refund retry and	Passed

07		refund with different permission sets.	authority can invoke either action.	manual refund actions were allowed only for staff with refund authority.	
WB-S-08	Payout settlement permission	Attempt payout settlement with and without settlement permission.	Only authorized staff can settle or reject a payout.	Authorized staff could settle or reject payouts; staff without permission were denied.	Passed

D. Black-Box Test Cases — Client

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
BB-C-01	Order fixed-price service	Browse → listing → requirements → create → pay.	Frozen terms display and next action becomes available.	The order displayed the agreed terms, and the next eligible action became available after payment.	Passed
BB-C-02	Accept negotiated offer	Chat → receive offer → accept → pay.	One order uses the offer values.	Accepting the offer created one order using the negotiated terms.	Passed
BB-C-03	Complete and review	Accept submitted order → enter rating.	Net release and one genuine review appear.	Completion released the net amount and allowed one completed-order review.	Passed
BB-C-04	Open a dispute	Open a paid active order, select Dispute, provide a reason, and confirm.	The dispute enters the staff queue and both participants see the appropriate state.	The dispute appeared in the staff queue, and both participants saw the disputed state.	Passed
BB-C-05	Empty marketplace search	Search for a term with no matching published listing.	A clear empty state appears without a runtime error.	The unmatched search displayed a clear empty state without a runtime error.	Passed
BB-C-06	Cancel before fulfillment	Open an eligible unpaid or pending order and choose Cancel with a	The order follows the legal	The eligible order followed the permitted	Passed

		reason.	cancellation/refund path.	cancellation and applicable refund path.	
BB-C-07	Notification deep link	Tap a client order or chat notification.	The relevant target opens and invalid links fail safely.	Valid notifications opened the relevant order or chat; invalid destinations failed safely.	Passed
BB-C-08	Open an active buyer order	Orders â†’ Buying â†’ select a pending active order.	The complete order-detail screen opens with status, terms, requirements, delivery, and payment information.	The pending buyer order 'Debug my code (Python or Dart)' opened successfully with its lifecycle tracker, frozen terms, requirements, delivery section, and unpaid payment summary; no error screen appeared.	Passed

E. Black-Box Test Cases — Freelancer

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
BB-F-01	Publish service	My Services → Add → valid fields → Publish.	Listing appears in marketplace search.	The published service appeared in marketplace search results.	Passed
BB-F-02	Fulfill order	Accept → start → submit delivery evidence.	Client sees submitted state and choices.	Submitted delivery evidence appeared for the client with the submitted status and available actions.	Passed
BB-F-03	Request payout	Wallet → configure account → request payout.	Request appears with expected queue state.	The payout request was created and appeared with the expected queue status.	Passed
BB-F-04	Pause and resume a	Open My Services, pause a	The service	The paused listing	Passed

	listing	published service, then publish it again.	disappears from search while paused and returns only when published.	disappeared from search and returned after it was published again.	
BB-F-05	Send an offer	Open a client conversation, create a valid offer, and submit it.	The client receives a readable offer card with scope, amount, revisions, and duration.	The client received an offer card showing scope, price, revisions, and delivery duration.	Passed
BB-F-06	Receive a client message	Client starts a conversation from a listing; freelancer opens Messages.	The conversation appears with the latest message and unread state.	The conversation appeared with the latest client message and unread state.	Passed
BB-F-07	Revision loop	Client requests revision on submitted work; freelancer opens the order.	The order returns to the permitted work state and can be resubmitted.	The revision request returned the order to the permitted work state and allowed resubmission.	Passed
BB-F-08	Clearing balance explanation	Complete a new seller's early order and open Wallet.	The amount appears as clearing rather than incorrectly available.	The early-order earnings appeared as clearing funds rather than available funds.	Passed
BB-F-09	No selling orders state	Orders â†’ Selling for a freelancer account that has no received orders.	A clear empty state and My services action appear without an error or indefinite loader.	The Selling tab loaded the intentional 'No orders for your services yet' state and displayed the My services action without an error or endless loading indicator.	Passed

F. Black-Box Test Cases — Authorized Staff

Test ID	Scenario	Preconditions and steps	Expected result	Actual result	Status
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BB-S-01	Moderate from profile	Students → profile → listing → take down.	Listing leaves public marketplace; action recorded.	The listing was removed from the public marketplace and the moderation action was recorded.	Passed
BB-S-02	Decide verification	Verification → inspect → approve/reject with note.	Status, notice, and audit update.	The verification decision updated the request status, notification, and audit record.	Passed
BB-S-03	Sales and Excel export	Overview → Sales → range → export.	Settled sales filter and XLSX has readable names.	The selected range filtered settled sales, and the Excel export displayed readable names.	Passed
BB-S-04	Open audit-entry details	Admin → Audit Log → select an entry with a linked subject and actor.	A readable detail screen opens without endless loading or raw document ids.	Audit details opened with readable subject and actor information without an endless loader.	Passed
BB-S-05	Filter and export sales	Select All time, Last 7 days, and Last 30 days; export the current view.	Each range changes the settled-sales result and export produces the matching XLSX report.	Each date filter updated the sales results, and the Excel export matched the selected view.	Passed
BB-S-06	Settle payout	Open an eligible payout and record a valid staff settlement.	The payout, wallet ledger, notification, and audit record update together.	Settlement updated the payout, wallet ledger, notification, and audit record together.	Passed
BB-S-07	Retry a stuck refund	Open a failed refund queue item and invoke retry.	The retry result is shown without leaving the queue indefinitely loading.	The refund retry displayed its result without leaving the queue loading indefinitely.	Passed

BB-S-08	Student-profile listing moderation	Open a student profile that owns multiple listings and moderate one item.	Only the selected listing changes; the profile remains readable.	Only the selected listing changed; the student profile remained readable.	Passed
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